

A Mechanistic Model for Radiation-Induced Segregation Controlled Grain-Boundary Oxidation in Austenitic Stainless Steels

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1 Introduction

2 Model Formulation and Calibration

The model consists of two components: a radiation-induced segregation (RIS) model for the evolving GB chemistry and a reduced-dimensional oxidation model for oxygen transport, consumption, and oxide growth. The RIS-predicted Cr, Ni, Si, and Fe concentrations provide the local alloy composition for the oxidation model. Oxygen transport along the GB is solved explicitly, while transport through the oxide and interfacial reactions are treated using a quasi-steady resistance formulation. The oxide composition determines the effective oxygen diffusivity through an effective-medium treatment, which controls the local oxygen consumption and oxide growth rate. The formulations of the two components are described below.

2.1 Radiation-Induced Segregation Model

The RIS model is based on solving a set of coupled mass-conservation equations for the alloying elements (Cr, Ni, Si, and Fe) and point defects (vacancies and self-interstitials). For each alloying element i , where $i \in \{\text{Cr, Ni, Si, Fe}\}$, the concentration evolves according to

$$\frac{\partial C_i}{\partial t} = -\nabla \cdot J_i. \quad (1)$$

The site fractions of vacancies and self-interstitials are governed separately by

$$\frac{\partial C_{\text{Vac}}}{\partial t} = -\nabla \cdot \mathbf{J}_{\text{Vac}} + \epsilon K_0 - RC_{\text{Int}}C_{\text{Vac}}, \quad (2)$$

$$\frac{\partial C_{\text{Int}}}{\partial t} = -\nabla \cdot \mathbf{J}_{\text{Int}} + \epsilon K_0 - RC_{\text{Int}}C_{\text{Vac}}, \quad (3)$$

where C_{Vac} and C_{Int} are the vacancy and self-interstitial site fractions, respectively, ϵ is the damage efficiency, K_0 is the radiation dose rate, and R is the vacancy–interstitial recombination rate coefficient. The flux of each species is given by

$$\mathbf{J}_i^{\text{Vac}} = -\sum_j L_{ij}^{\text{Vac}} \nabla (\mu_j - \mu_{\text{Vac}}), \quad (4)$$

$$\mathbf{J}_i^{\text{Int}} = -\sum_j L_{ij}^{\text{Int}} \nabla (\mu_j + \mu_{\text{Int}}), \quad (5)$$

$$\begin{aligned} L_{ii}^{\text{Vac}} &= \left(\frac{C_i D_i^{\text{Vac}} C_{\text{Vac}}}{k_B T} \right) \left(1 + \frac{1 - f_0^{\text{Vac}}}{f_0^{\text{Vac}}} \frac{C_i D_i^{\text{Vac}}}{\sum_m C_m D_m^{\text{Vac}}} \right), & i = j, \\ L_{ij}^{\text{Vac}} = L_{ji}^{\text{Vac}} &= \frac{(1 - f_0^{\text{Vac}})}{f_0^{\text{Vac}}} \frac{C_i D_i^{\text{Vac}} C_j D_j^{\text{Vac}} C_{\text{Vac}}}{k_B T \sum_m C_m D_m^{\text{Vac}}}, & i \neq j. \end{aligned} \quad (6)$$

$$\begin{aligned} L_{ii}^{\text{Int}} &= \left(\frac{C_i D_i^{\text{Int}} C_{\text{Int}}}{k_B T} \right) \left(1 + \frac{1 - f_0^{\text{Int}}}{f_0^{\text{Int}}} \frac{C_i D_i^{\text{Int}}}{\sum_m C_m D_m^{\text{Int}}} \right), & i = j, \\ L_{ij}^{\text{Int}} = L_{ji}^{\text{Int}} &= \frac{(1 - f_0^{\text{Int}})}{f_0^{\text{Int}}} \frac{C_i D_i^{\text{Int}} C_j D_j^{\text{Int}} C_{\text{Int}}}{k_B T \sum_m C_m D_m^{\text{Int}}}, & i \neq j. \end{aligned} \quad (7)$$

$$\mathbf{J}_{\text{Vac}} = -\sum_i \mathbf{J}_i^{\text{Vac}} \quad (8)$$

$$\mathbf{J}_{\text{Int}} = \sum_i \mathbf{J}_i^{\text{Int}} \quad (9)$$

$$v_{\text{lattice}}^{\text{RIS}} = \mathbf{J}_{\text{Vac}} - \mathbf{J}_{\text{Int}} \quad (10)$$

2.2 Grain-Boundary Oxidation Model

$$\frac{\partial C_O}{\partial t} = -\frac{\partial \mathbf{J}_O}{\partial y} - \frac{\mathbf{J}_r}{\delta_{GB}} \quad (11)$$

$$\mathbf{J}_O = -D_O^{\text{eff}} \frac{\partial C_O}{\partial y} \quad (12)$$

$$D_O^{\text{eff}} = \sum_i f_i D_O^i \quad (13)$$

$$\mathbf{J}_r = D_O^{\text{eff}} \frac{C_O - C_O^{\text{interface}}}{\sum_i \text{thick}_i} \quad (14)$$

$$\mathbf{J}_r^{\text{Cr}_2\text{O}_3} = k_r^{\text{Cr}_2\text{O}_3} C_O^{\text{interface}} C_{\text{Cr}}^{\text{GB}} \quad (15)$$

$$\mathbf{J}_r^{\text{Fe}_3\text{O}_4} = k_r^{\text{Fe}_3\text{O}_4} (C_O^{\text{interface}} - C_O^{\text{threshold}}) C_{\text{Fe}}^{\text{GB}} \quad (16)$$

$$\mathbf{J}_r^{\text{FeCr}_2\text{O}_4} = k_r^{\text{FeCr}_2\text{O}_4} C_O^{\text{interface}} \frac{1}{\frac{1}{C_{\text{Fe}}^{\text{GB}}} + \frac{2}{C_{\text{Cr}}^{\text{GB}}}} \quad (17)$$

$$\mathbf{J}_r^{\text{SiO}_2} = k_r^{\text{SiO}_2} C_O^{\text{interface}} C_{\text{Si}}^{\text{GB}} \quad (18)$$

$$\mathbf{J}_r = \mathbf{J}_r^{\text{Cr}_2\text{O}_3} + \mathbf{J}_r^{\text{Fe}_3\text{O}_4} + \mathbf{J}_r^{\text{FeCr}_2\text{O}_4} + \mathbf{J}_r^{\text{SiO}_2} \quad (19)$$

$$\mathbf{J}_{\text{Cr}} = \frac{2}{3} \mathbf{J}_r^{\text{Cr}_2\text{O}_3} + \frac{2}{4} \mathbf{J}_r^{\text{FeCr}_2\text{O}_4} \quad (20)$$

$$\mathbf{J}_{\text{Fe}} = \frac{3}{4} \mathbf{J}_r^{\text{Fe}_3\text{O}_4} + \frac{1}{4} \mathbf{J}_r^{\text{FeCr}_2\text{O}_4} \quad (21)$$

$$\mathbf{J}_{\text{Si}} = \frac{1}{2} \mathbf{J}_r^{\text{SiO}_2} \quad (22)$$

2.3 Coupling Strategy and Simulation Protocol

2.3.1 Lattice Velocity and Metal Consumption

$$v_{\text{lattice}}^{\text{reaction}} = \mathbf{J}_{\text{r}}^{\text{Cr}} + \mathbf{J}_{\text{r}}^{\text{Fe}} + \mathbf{J}_{\text{r}}^{\text{Si}} \quad (23)$$

$$v_{\text{lattice}} = v_{\text{lattice}}^{\text{RIS}} + v_{\text{lattice}}^{\text{reaction}} \quad (24)$$

$$\mathbf{J}_i = \mathbf{J}_i^{\text{Vac}} + \mathbf{J}_i^{\text{Int}} + \mathbf{J}_{\text{r}}^i + C_i v_{\text{lattice}} \quad (25)$$

- 2.3.2 Sequential Irradiation–Oxidation Protocol
- 2.3.3 Fully Coupled Irradiation–Oxidation Framework
- 2.4 Model Parameterization and Calibration
 - 2.4.1 RIS Parameterization
 - 2.4.2 Oxidation Kinetics Parameterization

3 Results

- 3.1 Radiation-Induced Segregation Profiles
- 3.2 Dose-Dependent Grain-Boundary Oxidation Kinetics
- 3.3 Oxide Composition and Interfacial Ni Enrichment
- 3.4 Predictions against Independent Experiments
- 3.5 Fully Coupled Predictions under Concurrent Irradiation and Oxidation

4 Discussion

- 4.1 Effect of Radiation-Induced Segregation on Grain-Boundary Oxidation
- 4.2 Rate-Controlling Transport and Oxide Chemistry
- 4.3 Implications of Concurrent Irradiation and Oxidation
- 4.4 Predictive Capability and Model Limitations

5 Conclusions